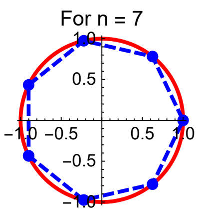
Practical 1

Ques nth root of unity geometric plot

In[•]:=

```
n = 7;
roots = ComplexExpand[z /. N[Solve[z^n - 1 == 0]]];
pts = {Re[#], Im[#]} & /@ roots;
pts = SortBy[pts, ArcTan @@ # &];
A1 = Show[Graphics[{Red, Thick, Circle[{0, 0}, 1]}],
   ListLinePlot[Append[pts, First[pts]],
    PlotStyle → {Blue, Dashed, Thick}, PlotMarkers → Automatic],
   Axes → True, ImageSize → 100, PlotLabel → "For n = 7"]
```

Out[•]=

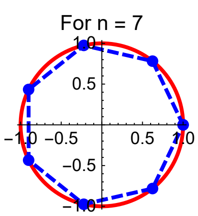


or

In[•]:=

```
n = 7;
roots = Exp[2 Pi I Range[0, n - 1] / n];
pts = {Re[#], Im[#]} & /@ roots;
A1 = Show[Graphics[{Red, Thick, Circle[{0, 0}, 1]}],
   ListLinePlot[Append[pts, First[pts]],
    PlotStyle → {Blue, Dashed, Thick}, PlotMarkers → Automatic],
   Axes → True, ImageSize → 100, PlotLabel → "For n = 7"]
```

Out[•]=

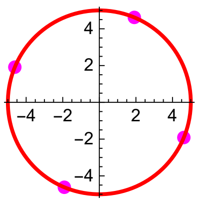


Practical 2

Find solution of z^4=-625*I and represent solution geometrically .

In[∘]:=
```
sol = ComplexExpand[z /. Solve[z^4 + 625 * I == 0]]
pts = {Re[#], Im[#]} & /@ sol;
a = Graphics[{Red, Thick, Circle[{0, 0}, 5]}];
b = Graphics[{Magenta, PointSize[Large], Point[pts]}, ImageSize → 100, Axes -> True];
Show[b, a]
```

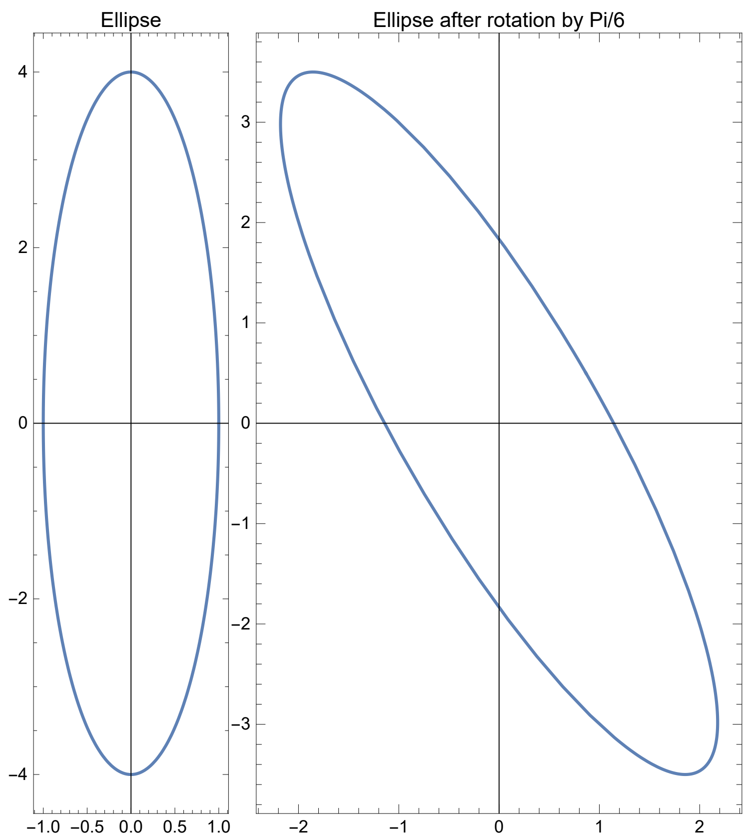
Out[∘]= $\left\{5\,\text{Cos}\left[\frac{\pi}{8}\right]-5\,i\,\text{Sin}\left[\frac{\pi}{8}\right],\ -5\,\text{Cos}\left[\frac{\pi}{8}\right]+5\,i\,\text{Sin}\left[\frac{\pi}{8}\right],\ 5\,i\,\text{Cos}\left[\frac{\pi}{8}\right]+5\,\text{Sin}\left[\frac{\pi}{8}\right],\ -5\,i\,\text{Cos}\left[\frac{\pi}{8}\right]-5\,\text{Sin}\left[\frac{\pi}{8}\right]\right\}$

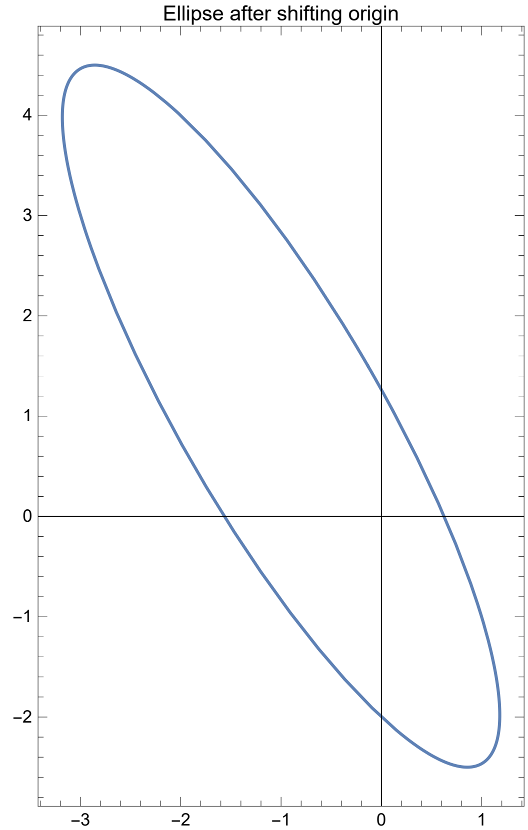
Out[∘]=


Practical 3

Ques effect of rotation of ellipse by pi/6 radians and centre shifting from origin to (-1,1) using parametric plot, eqn is (cost,4sint)

In[∘]:=
```
A1 = ParametricPlot[{Cos[t], 4 Sin[t]}, {t, 0, 2 Pi},
   Frame → True, ImageSize → Medium, PlotLabel → "Ellipse"];
A2 = ParametricPlot[{Cos[t] Cos[Pi / 6] - 4 Sin[t] Sin[Pi / 6],
    Cos[t] Sin[Pi / 6] + 4 Sin[t] Cos[Pi / 6]}, {t, 0, 2 Pi}, Frame → True,
   ImageSize → Medium, PlotLabel → "Ellipse after rotation by Pi/6"];
A3 = ParametricPlot[{Cos[t] Cos[Pi / 6] - 4 Sin[t] Sin[Pi / 6] - 1,
    Cos[t] Sin[Pi / 6] + 4 Sin[t] Cos[Pi / 6] + 1}, {t, 0, 2 Pi}, Frame → True,
   ImageSize → Medium, PlotLabel → "Ellipse after shifting origin"];
Print[
 A1,
 A2,
 A3]
```


Ellipse
Ellipse after rotation by Pi/6
4
2
0
−2
−4
−1.0
−0.5
0.0
0.5
1.0
3
2
1
0
−1
−2
−3
−2
−1
0
1
2



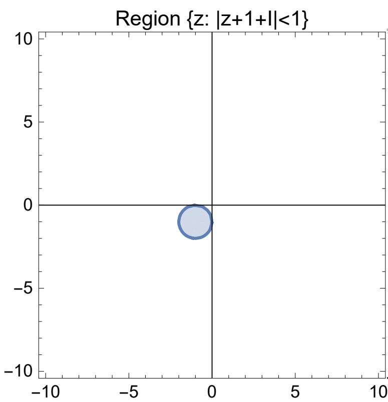
Practical 4

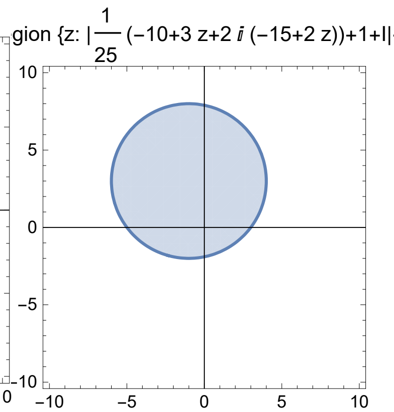
Show that the image of the open disk D1 (−1 − □) = {□ : □ + 1 + □ < 1} under the linear transformation w = f (z) = (3−4 i) z + (6 + 2 i) is the open disk : D5 (−1 + 3 i) = {w : w + 1−3 i < 5}.

In[ ]:=
```
Solve[w1 == (3 - 4 I) z1 + (6 + 2 I), z1]
```

Out[ ]=
$$\left\{\left\{z1 \to \frac{1}{25}\left(-10 + 3\,w1 + 2\,i\,(-15 + 2\,w1)\right)\right\}\right\}$$

In[ ]:=
```
z = x + I * y;
A1 = RegionPlot[Abs[z + 1 + I] < 1, {x, -10, 10}, {y, -10, 10},
   Axes → True, ImageSize → 200, PlotLabel → "Region {z: |z+1+I|<1}"];
A2 = RegionPlot[Abs[1/25 (-10 + 3 z + 2 I (-15 + 2 z)) + 1 + I] < 1, {x, -10, 10}, {y, -10, 10},
   Axes → True, ImageSize → 200,
   PlotLabel → "Region {z: |1/25 (-10+3 z+2 ⅈ (-15+2 z))+1+I|<1}"];
Print[A1, A2]
```



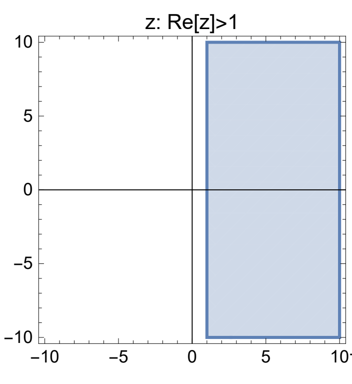


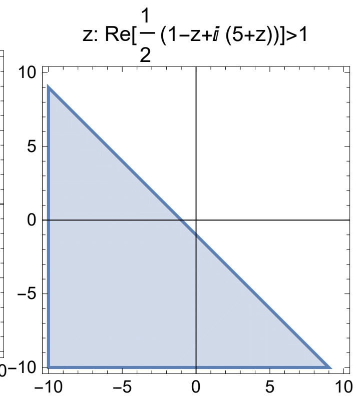
Practical 5

In[ ]:=
```
Solve[w1 == (-1 - I) z1 - 2 + 3 I, z1]
```

Out[ ]=
$$\left\{\left\{z1 \to \frac{1}{2}\left(1 - w1 + i\left(5 + w1\right)\right)\right\}\right\}$$

In[ ]:=
```
z = x + I * y;
A1 = RegionPlot[Re[z] > 1, {x, -10, 10}, {y, -10, 10},
   Axes → True, ImageSize → Small, PlotLabel → "z: Re[z]>1"];
A2 = RegionPlot[Re[1/2 (1 - z + i (5 + z))] > 1, {x, -10, 10}, {y, -10, 10},
   Axes → True, ImageSize → Small, PlotLabel → "z: Re[1/2 (1-z+i (5+z))]>1"];
Print[
 A1,
 A2]
```



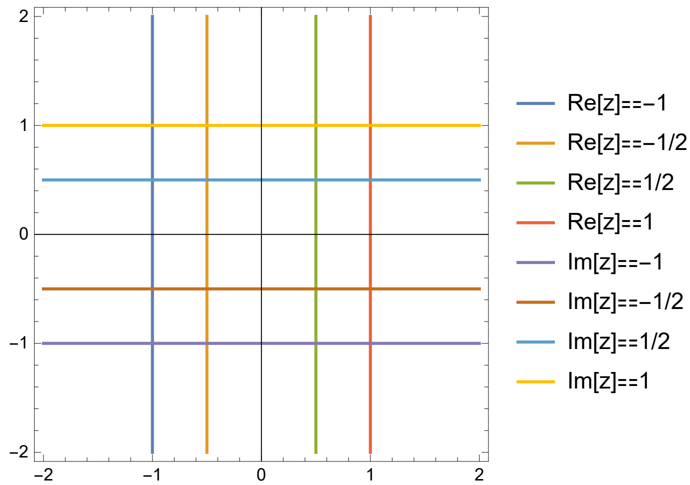


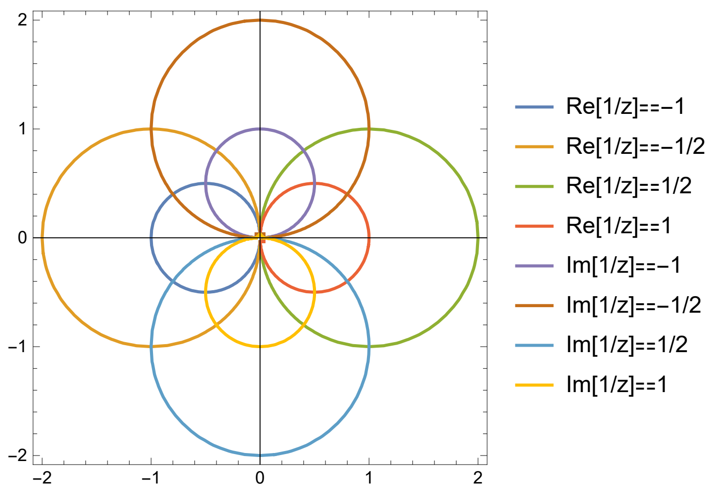
Practical 6
Same as 4 & 5

Practical 7

Make a plot of the vertical lines x=a for a = -1, -1/2, 1/2, 1 and horizontal lines y = b for b = -1, -1/2, 1/2, 1. Find plot of this grid under mapping w = f[z]= 1/z.

```
In[ ]:= z = x + I * y;
A1 = ContourPlot[{Re[z] == -1, Re[z] == -1/2, Re[z] == 1/2, Re[z] == 1, Im[z] == -1,
    Im[z] == -1/2, Im[z] == 1/2, Im[z] == 1}, {x, -2, 2}, {y, -2, 2}, Axes → True,
   ImageSize → 250, PlotLegends → {"Re[z]==-1", "Re[z]==-1/2", "Re[z]==1/2",
     "Re[z]==1", "Im[z]==-1", "Im[z]==-1/2", "Im[z]==1/2", "Im[z]==1"}];
A2 = ContourPlot[{Re[1/z] == -1, Re[1/z] == -1/2, Re[1/z] == 1/2, Re[1/z] == 1,
    Im[1/z] == -1, Im[1/z] == -1/2, Im[1/z] == 1/2, Im[1/z] == 1},
   {x, -2, 2}, {y, -2, 2}, Axes → True, ImageSize → 250,
   PlotLegends → {"Re[1/z]==-1", "Re[1/z]==-1/2", "Re[1/z]==1/2", "Re[1/z]==1",
     "Im[1/z]==-1", "Im[1/z]==-1/2", "Im[1/z]==1/2", "Im[1/z]==1"}];
Print[
 A1,
 A2]
```


Re[z]==−1
Re[z]==−1/2
Re[z]==1/2
Re[z]==1
Im[z]==−1
Im[z]==−1/2
Im[z]==1/2
Im[z]==1


Re[1/z]==−1
Re[1/z]==−1/2
Re[1/z]==1/2
Re[1/z]==1
Im[1/z]==−1
Im[1/z]==−1/2
Im[1/z]==1/2
Im[1/z]==1

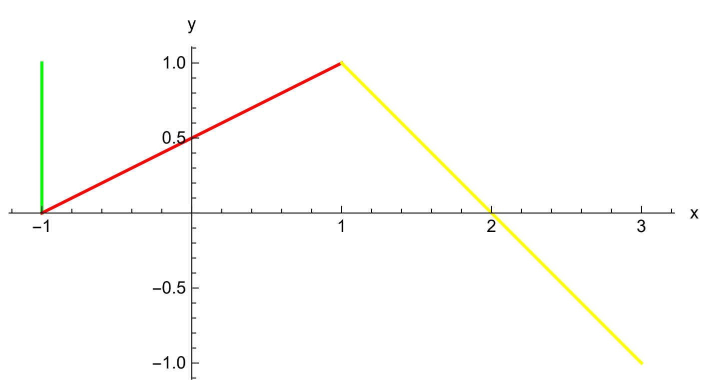
Practical 8

Find a parametric Representation of the path C = C1 + C2 + C3, FROM - 1 + I to 3 - I, where C1 is the line from - 1 + I to - 1; C2 is a line from - 1 to 1 + I and C 3 is a line from 1 + I to 3 - I. Make a plot of this path.

In[◦]:=
```
w1 := -1 + I;
w2 := -1;
w3 := 1 + I;
w4 := 3 - I;
z1[t_] := (1 - t) w1 + t w2;
z2[t_] := (1 - t) w2 + t w3;
z3[t_] := (1 - t) w3 + t w4;
Print["The Parametric representation of path C1,C2,C3 is given by"]
Print["C1: z1[t]=", z1[t], "  0<=t<=1"]
Print["C2: z2[t]=", z2[t], "  0<=t<=1"]
Print["C3: z3[t]=", z3[t], "  0<=t<=1"]
Print["The plot of the polynomial C= C1+C2+C3 is given by figure "]
ParametricPlot[
 {{Re[z1[t]], Im[z1[t]]}, {Re[z2[t]], Im[z2[t]]}, {Re[z3[t]], Im[z3[t]]}},
 {t, 0, 1}, PlotStyle → {Green, Red, Yellow}, AxesLabel → {"x", "y"}]
```

```
The Parametric representation of path C1,C2,C3 is given by

C1: z1[t]=(-1 + i) (1 - t) - t  0<=t<=1

C2: z2[t]=-1 + (2 + i) t  0<=t<=1

C3: z3[t]=(1 + i) (1 - t) + (3 - i) t  0<=t<=1

The plot of the polynomial C= C1+C2+C3 is given by figure
```

Out[◦]=



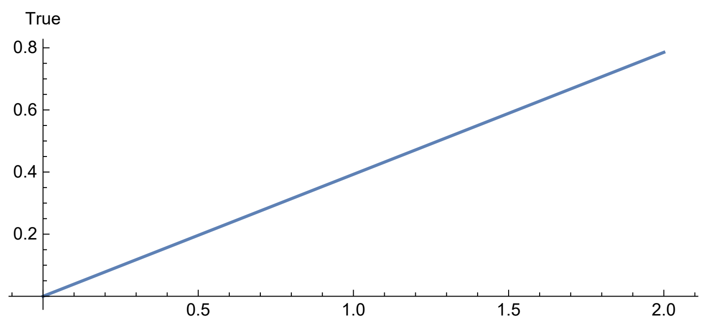
Practical 9

Find the line segment “L” joining the point A=0 to B=2+I (Pi/4), and give the exact calculation of Contour Integral e^z dz

In[•]:=
```
a := 0;
b := 2 + I * Pi / 4;
z[t_] := (1 - t) a + t * b;
f[z_] := Exp[z];
Print[
  " The parametrization of line seqment l joining points A to B is ", z[t], "   0≤t≤1"];
Print["The plot of line segment L is given by figure"];
ParametricPlot[{Re[z[t]], Im[z[t]]}, {t, 0, 1}, AxesLabel → True]
Print["Value of Integral is ", ComplexExpand[Integrate[f[z[t]] * z'[t], {t, 0, 1}]]]
```

The parametrization of line seqment l joining points A to B is $\left(2+\frac{i\pi}{4}\right)t \quad 0\leq t\leq 1$

The plot of line segment L is given by figure

Out[•]=


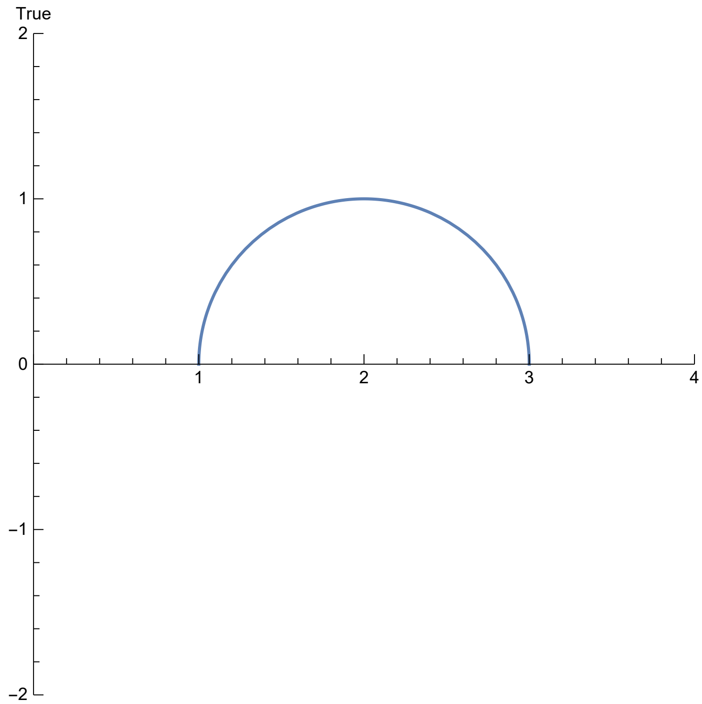
Value of Integral is $-1+\frac{(1+i)\,e^2}{\sqrt{2}}$

Practical 10

Plot the semicircle’C’with the radius 1 at z = 2and evaluate the contour integral ∫c 1/(z-2)dz

In[•]:=
```
Quit[]
```

In[]:=
```
z[t_] := 2 + Cos[t] + I * Sin[t];
f[z_] := 1/(z - 2);
ParametricPlot[{Re[z[t]], Im[z[t]]},
 {t, 0, Pi}, PlotRange → {{0, 4}, {-2, 2}}, AxesLabel → True]
Print["Value of Integral is ", ComplexExpand[Integrate[f[z[t]] * z'[t], {t, 0, Pi}]]]
```

Out[]=


Value of Integral is $i\pi$

Practical 11

Show that $\int$c1 zdz = $\int$c2 zdz = 4 + 2*I, where C1 is the line segment from-1+I to 3+I, andC2 is the portion of parabola x = y^2 +2y joining -I to 3+I.Make plots of two contours C1 and C2 joining -1-I to 3+I.

In[ ]:=
```
w1 := -1 - I;
w2 := 3 + I;
z1[t_] := (1 - t) * w1 + t * w2;
z2[t_] := (t^2 + 2 t) + I * t;
Print["The parametrization of the line segment C1 is given by"]
Print["C1: z[t] = ", z1[t], " , 0<=t<=1"]
Print["The parametrization of the portion of parabola C2 is given by"]
Print["C2: z[t] = ", z2[t], " ,-1<=t<=1"]
Print["The plots of the line segment
   C1 and portion of the parabola C2 is given by the figure"]
Show[ParametricPlot[{Re[z1[t]], Im[z1[t]]}, {t, 0, 1}, PlotStyle → {Thick, Red}],
 ParametricPlot[{Re[z2[t]], Im[z2[t]]}, {t, -1, 1}, PlotStyle → {Thick, Blue}]]
f[z_] := z;
Print["∫c1 zdz = ", Integrate[f[z1[t]] * z1'[t], {t, 0, 1}], "------(1)"]
Print["∫c2 zdz = ", Integrate[f[z2[t]] * z2'[t], {t, -1, 1}], "------(2)"]
Print["There by (1) and (2), we proved the required part."]
```

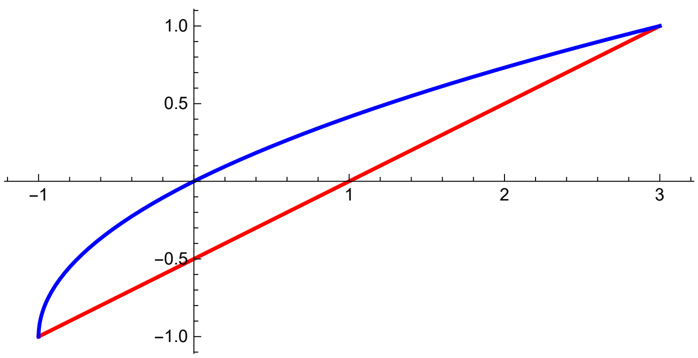
The parametrization of the line segment C1 is given by

C1: z[t] = $(-1 - i)(1 - t) + (3 + i)t$, 0<=t<=1

The parametrization of the portion of parabola C2 is given by

C2: z[t] = $(2 + i)t + t^2$,-1<=t<=1

The plots of the line segment C1 and portion of the parabola C2 is given by the figure

Out[ ]=


∫c1 zdz = 4 + 2 i------(1)

∫c2 zdz = 4 + 2 i------(2)

There by (1) and (2), we proved the required part.

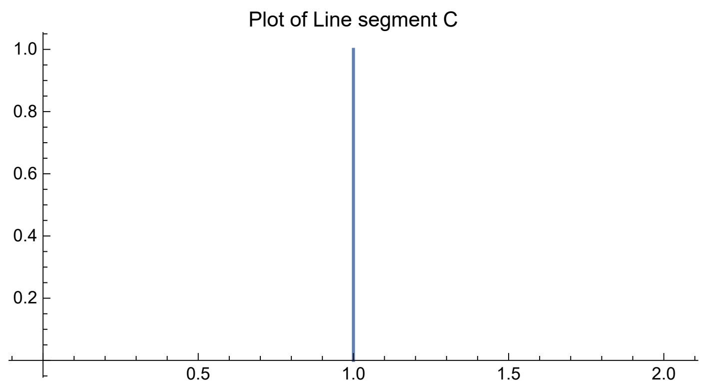
Practical 12

In[ ]:=
```
Quit[]
```

In[∘]:=
```
a := 1;
b := 1 + I;
z[t_] := (1 - t) a + t * b;
f[z_] := z^2;
Print["The parametrization of the line segment C : ", z[t]];
ParametricPlot[{Re[z[t]], Im[z[t]]}, {t, 0, 1}, PlotLabel → "Plot of Line segment C"]
L = Integrate[Abs[z'[t]], {t, 0, 1}];
r = ComplexExpand[Re[f[z[t]]]];
i = ComplexExpand[Im[f[z[t]]]];
M = Maximize[{(r^2 + i^2)^(1/2), 0 ≤ t ≤ 1}, t];
ML = M[[1]] * L;
Print["Thus by the ML inequality, upper bound of given integral is ", ML, "<=2√2 "]
Print["Hence the required inequality obtained"]
```

The parametrization of the line segment C : $1 + i\, t$

Out[∘]=



Thus by the ML inequality, upper bound of given integral is $2 <= 2\sqrt{2}$

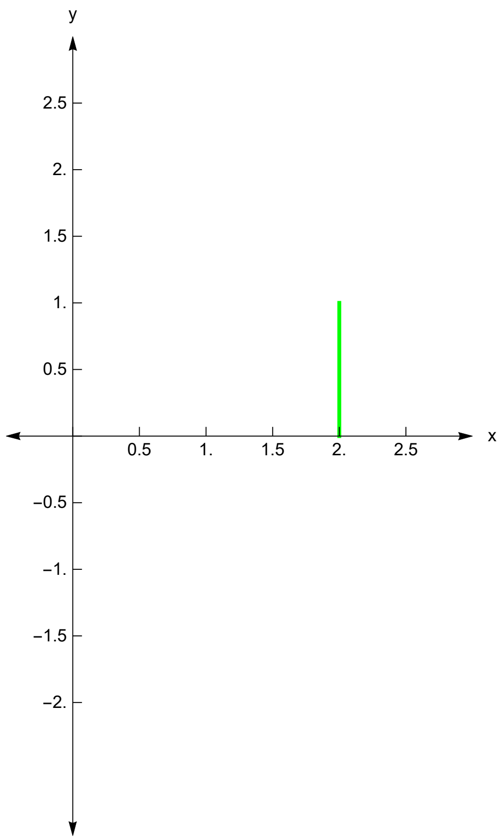
Hence the required inequality obtained

Or

In[ ]:=
```
z0 = 2;
z1 = 2 + I;
z[t_] = z0 + t * (z1 - z0);
Print["The parametrization equation of the given curve is C : z[t] = ",
 ComplexExpand[z[t]],
 " Where 0<t⩽1, which represent straight line joining the points z0 = ", z0,
 " and z1 = ", z1, "."]
p1 = ParametricPlot[{Re[z[t]], Im[z[t]]}, {t, 0, 1}, AxesStyle → Arrowheads[{-0.03, 0.03}],
   AxesLabel → {"x", "y"}, PlotStyle → {Thick, Green}, AxesOrigin → {0, 0},
   Ticks → {Range[-1, 3, .5], Range[-2, 2.5, 0.5]}, PlotRange → {{-0.5, 3}, {-3, 3}}];
Show[p1]
L = Integrate[Abs[z'[t]], {t, 0, 1}];
f[z_] = 1/(z^2 + 1);
r = ComplexExpand[Re[f[z[t]]]];
i = ComplexExpand[Im[f[z[t]]]];
M = Maximize[{(r^2 + i^2)^(1/2), 0 ≤ t ≤ 1}, t];
ML = M[[1]] * L;
Print["Length of the Arc is, L = ", 1, " and the value of M is ", M[[1]], "."]
Print[
 "Using the ML Inequality, we have an upper bound for the given integral as", ML, "."]
Manipulate[Show[ParametricPlot[{Re[z[t]], Im[z[t]]},
    {t, 0, 1}, PlotStyle → {Thick, Red}, PlotRange → 2],
  ListLinePlot[{{{0, 1}, {Re[z[t]], Im[z[t]]}}, {{0, -1}, {Re[z[t]], Im[z[t]]}}},
   AxesOrigin → {0, 0}]], {t, 0, 1}]
```

```
The parametrization equation of the given curve is C : z[t] = 2 + i t
  Where 0<t⩽1, which represent straight line joining the points z0 = 2 and z1 = 2 + i.
```

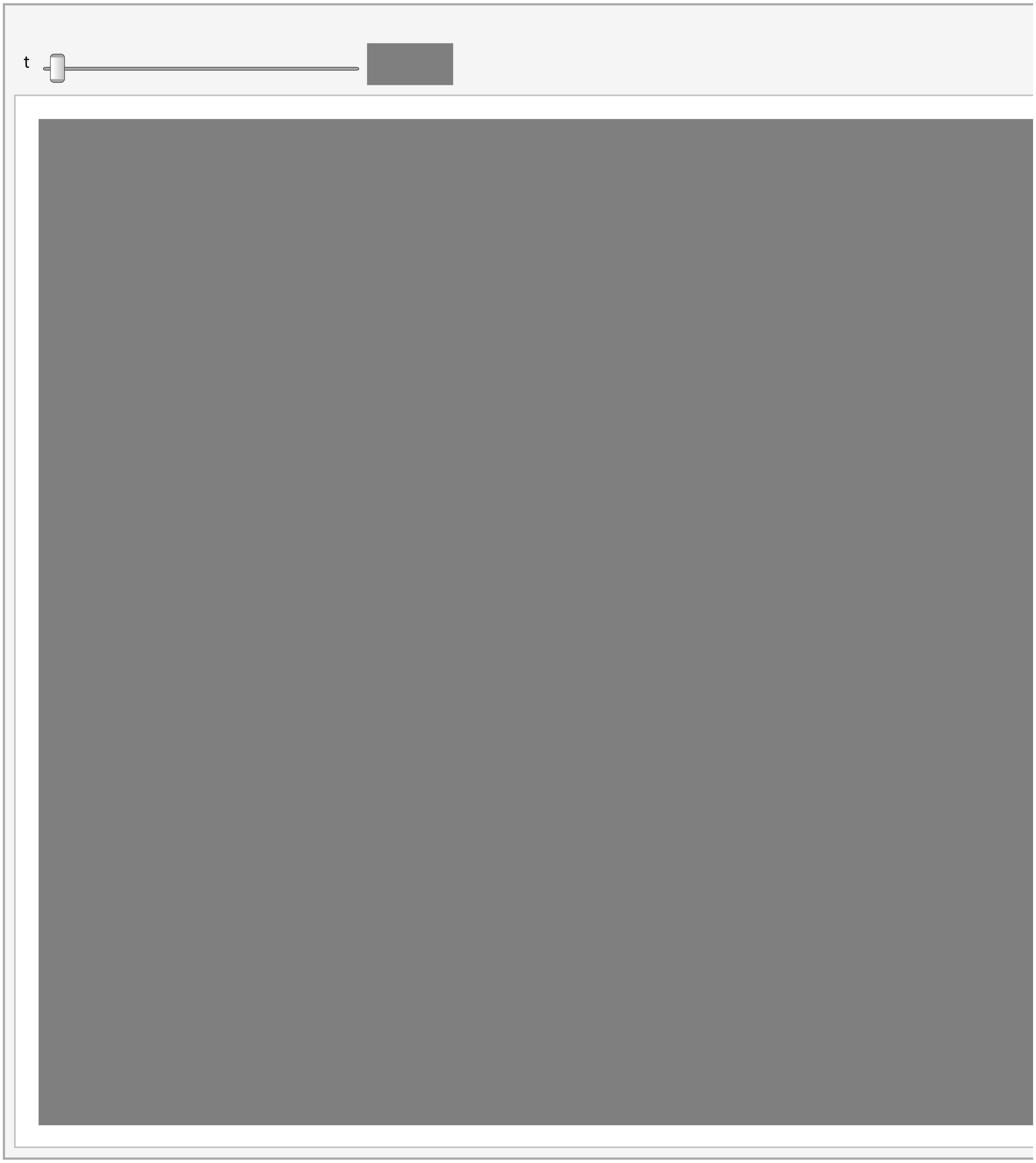
Out[ ]=



Length of the Arc is, L = 1 and the value of M is $\frac{1}{5}$.

Using the ML Inequality, we have an upper bound for the given integral as $\frac{1}{5}$.

Out[ ]=



Practical 13
Find and plot three Laurant Series representation for the function f[z]= 3/(2+z- z^2) involving powers of z.

In[ ]:=
```
f[z_] := 3/(2 + z - z^2);
s1 = Normal[Series[f[z], {z, 0, 10}]];
s2 = Normal[Series[f[z], {z, 2, 10}]];
s3 = Normal[Series[f[z], {z, Infinity, 10}]];
Print["Laurent Series around z=0 : ", s1];
Print["Laurent Series around z=2 : ", s2];
Print["Laurent Series around z=Infinity : ", s3];
Plot[{s1, s2, s3}, {z, -3, 3}, PlotLegends → {"About z=0", "About z=2", "z=Infinity"}]
```

Laurent Series around z=0 :

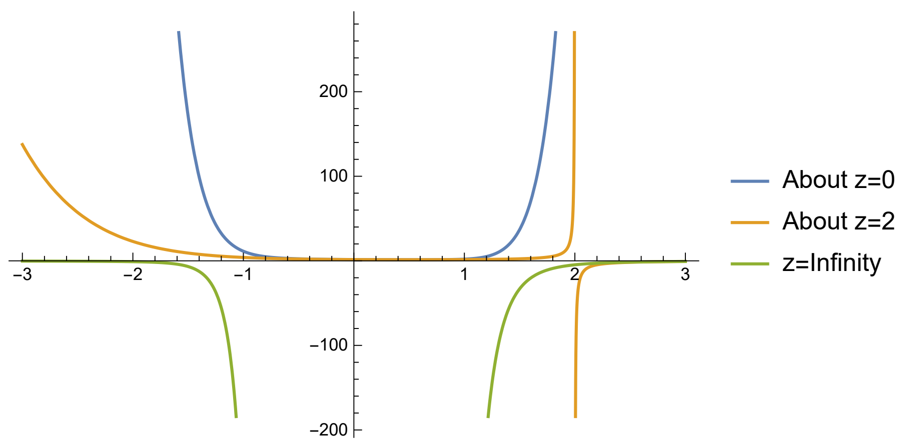
$$\frac{3}{2}-\frac{3z}{4}+\frac{9z^2}{8}-\frac{15z^3}{16}+\frac{33z^4}{32}-\frac{63z^5}{64}+\frac{129z^6}{128}-\frac{255z^7}{256}+\frac{513z^8}{512}-\frac{1023z^9}{1024}+\frac{2049z^{10}}{2048}$$

Laurent Series around z=2 : $\frac{1}{3}+\frac{2-z}{9}-\frac{1}{-2+z}+\frac{1}{27}(-2+z)^2-\frac{1}{81}(-2+z)^3+$

$$\frac{1}{243}(-2+z)^4-\frac{1}{729}(-2+z)^5+\frac{(-2+z)^6}{2187}-\frac{(-2+z)^7}{6561}+\frac{(-2+z)^8}{19683}-\frac{(-2+z)^9}{59049}+\frac{(-2+z)^{10}}{177147}$$

Laurent Series around z=Infinity : $-\frac{513}{z^{10}}-\frac{255}{z^9}-\frac{129}{z^8}-\frac{63}{z^7}-\frac{33}{z^6}-\frac{15}{z^5}-\frac{9}{z^4}-\frac{3}{z^3}-\frac{3}{z^2}$

Out[ ]=



Practical 14
Locate the poles of f[z] = 1/(5z^4 + 26 z^2 + 5)

```
In[•]:= f[z_] := 1/(5 z^4 + 26 z^2 + 5);
sol = Solve[Denominator[f[z]] == 0, z];
a = z /. sol;
Print["Poles are z1=", a[[1]], ",z2=", a[[2]], ", z3=", a[[3]], " & z4 = ", a[[4]]]
```

Poles are z1= $-\frac{i}{\sqrt{5}}$,z2= $\frac{i}{\sqrt{5}}$, z3= $-i\sqrt{5}$ & z4 = $i\sqrt{5}$

Practical 15

Locate the poles and zeroes of g[z] = πCot[πz]/z^2, Determine their order, also justify that Res(g,θ)=--(π^2)/3

```
In[•]:= g[z_] := Pi * Cot[Pi * z]/z^2;
pole = Solve[Denominator[g[z]] == 0, z];
zero = Reduce[g[z] == 0, z];
res = Residue[g[z], {z, 0}];
Print["Poles are  ", pole];
Print["zeroes are   ", zero];
Print["Residue of g[z] at z=0 is ", res]
```

Poles are {{z → 0}, {z → 0}}

zeroes are $c_1 \in \mathbb{Z}$ && $z \neq 0$ && $z == \frac{\frac{\pi}{2} + \pi c_1}{\pi}$

Residue of g[z] at z=0 is $-\frac{\pi^2}{3}$

Practical 16

Evaluate $\int$c1(θ) e^(2/z) dz , where c1(0) denotes the circle {z : |z|=1} with positive orientation. Similarly evaluate $\int_{c1} \frac{1}{z^4+z^3-2z^2}\, dz$.

```
In[•]:= g[z_] := 1/(z^4 + z^3 - 2 z^2);
Apart[g[z]]
```

Out[•]= $\frac{1}{3(-1+z)} - \frac{1}{2z^2} - \frac{1}{4z} - \frac{1}{12(2+z)}$

```
In[ ]:= r1 = SeriesCoefficient[g[z], {z, 0, -1}];
        r2 = SeriesCoefficient[g[z], {z, 1, -1}];
        r3 = SeriesCoefficient[g[z], {z, -2, -1}];
        Print[" The residue of g[z] at z = 0 is ",
          r1, " , at z = 1 is ", r2, " & at z = -2 is ", r3];
        integral = 2 Pi I * (r1 + r2 + r3);
```

Print["The value of $\int_{c1} \frac{1}{z^4 + z^3 - 2 z^2} dz$ by Cauchy residue theorem is ", integral]

The residue of g[z] at z = 0 is $-\frac{1}{4}$, at z = 1 is $\frac{1}{3}$ & at z = −2 is $-\frac{1}{12}$

The value of $\int_{c1} \frac{1}{z^4 + z^3 - 2 z^2} dz$ by Cauchy residue theorem is 0